

# ON THE PITFALLS OF VERBALIZED CONFIDENCE PRIORS FOR CALIBRATING REASONING LLMs

**Anonymous authors**

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## ABSTRACT

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## 1 INTRODUCTION

Reasoning large language models (LLMs) increasingly solve complex problems by generating a reasoning trajectory followed by the answer (Yang et al., 2025). Despite these advances, the answer accuracy alone does not ensure reliable uncertainty estimates: a model may produce plausible reasoning trajectory yet remain overconfident in an incorrect answer (Geng et al., 2024; Liu et al., 2025). In principle, confidence calibration requires the reported confidence to reflect the empirical probability that the answer is correct, which in turn enables applications such as hallucination detection (Wu et al., 2026) and adaptive test-time scaling (Yang et al., 2026).

Verbalized confidence obtains such an answer-level score by asking the model to emit a numerical probability with its final answer. Unlike confidence estimators based on token probabilities, hidden states, or repeated sampling (Luo et al., 2026; Srey et al., 2026; Wang & Stengel-Eskin, 2026), it can be produced in a single black-box generation (Xiong et al., 2024). Recent work trains this output with supervised fine-tuning (SFT), using targets derived from correctness or rollout statistics (Lin et al., 2022; Li et al., 2025), and with reinforcement learning (RL), by adding calibration terms to answer rewards (Xu et al., 2024; Damani et al., 2026; Ma et al., 2026; Yang et al., 2026).

We identify one primary failure and one recipe-dependent optimization effect. Under a fixed prompt, format, and decoding protocol, the base model’s sampled confidence is concentrated on a small set of values and only moderately stratifies correct and incorrect rollouts. The resulting confidence prior provides limited support for learning a graded correctness estimate. In filtering ablations, completed no-filtering DCPO and RLCR runs further concentrate confidence on a few values and endpoints, whereas the corresponding DAPO-filtered runs retain broader support. We therefore treat confidence concentration as a property of the sampling recipe, rather than as an inevitable consequence of GRPO.

We address the initialization problem with CalibSFT. CalibSFT constructs a target for each rollout by combining the success rate of its prompt group with the rollout’s verified correctness. Its selective segment loss retains confidence supervision on incorrect rollouts while restricting reasoning/answer supervision to correct rollouts, and target-balanced sampling exposes the model to a broader range of confidence values. In the subsequent RL stage, all compared methods use the same DAPO-style mixed-outcome filtering (Yu et al., 2025) as a fixed sampling recipe, allowing us to compare Base and CalibSFT initialization under controlled downstream optimization.

We make three contributions:

- We characterize a concentrated base-model confidence prior and document how confidence concentration depends on the RL sampling recipe.
- We introduce CalibSFT, which combines rollout-derived targets, target-balanced sampling, and correctness-aware segment supervision for confidence initialization.
- We evaluate whether the CalibSFT initialization improves accuracy, calibration, and confidence diversity after downstream RL across multiple confidence-aware objectives.

## 2 PRELIMINARIES

### 2.1 VERBALIZED CONFIDENCE FOR REASONING LLMs

In this work, we study confidence calibration for reasoning language models that verbalize a numerical confidence score (Tian et al., 2023; Xiong et al., 2024). Given a question  $x$ , the LLM is tasked with producing a completion  $o = (r, a, c)$ , where  $r$  is a reasoning trace and  $a$  is an answer.  $c \in [0, 1]$  is a post-answer confidence value and represents the model’s estimated probability that  $a$  is correct.

**Reinforcement Learning with Verifiable Rewards (RLVR).** RLVR formulates the LLM as a policy  $\pi_\theta$  parameterized by  $\theta$ . Given a question  $x$ , the policy generates a rollout  $o$ , and a task verifier assigns it a binary correctness label  $z = V(x, a) \in \{0, 1\}$ . Typically, the policy is optimized using Group Relative Policy Optimization (GRPO) (Shao et al., 2024). In practice, GRPO samples  $G$  rollouts  $\{o_i\}_{i=1}^G$  for each question, and the optimization objective can then be formulated as:

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[ \frac{1}{G} \sum_{i=1}^G \frac{1}{L_i} \sum_{t=1}^{L_i} \min \left( \rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_i \right) \right]. \quad (1)$$

Here  $o_i = (o_{i,1}, \dots, o_{i,L_i})$ , and  $\rho_{i,t}(\theta) = \pi_\theta(o_{i,t} | x, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} | x, o_{i,<t})$  is the policy ratio;  $\epsilon_{\text{clip}}$  is the clipping parameter. Given the reward  $R_i = R(o_i)$  and a small constant  $\epsilon > 0$  for numerical stability, the group-relative advantage is:

$$\hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \quad \bar{R} = \frac{1}{G} \sum_{j=1}^G R_j, \quad \sigma_R = \text{std}(\{R_j\}_{j=1}^G). \quad (2)$$

**Confidence-Aware Rewards.** For verbalized confidence training, a confidence-based scoring term is typically added to the reward. For example, RLCR (Damani et al., 2026) adds the negative Brier loss to the binary correctness reward:

$$R_i = z_i - (c_i - z_i)^2, \quad (3)$$

where  $c_i$  denotes the reported confidence and  $z_i = V(x, a_i) \in \{0, 1\}$  denotes the correctness label of rollout  $i$ . This reward favors correct answers while assigning higher scores to confidence values that better match their correctness labels, thereby increasing their probability under GRPO.

### 2.2 CALIBRATION METRICS

A useful verbalized confidence score should be both calibrated and discriminative (Eusebi et al., 2026). To formalize calibration, let  $C$  and  $Z$  denote the confidence and correctness random variables, respectively. A perfectly calibrated score satisfies  $\Pr(Z = 1 | C = p) = p$  for every  $p \in [0, 1]$ . To quantify the degree of miscalibration, **Expected Calibration Error (ECE)** (Guo et al., 2017) is defined as  $\mathbb{E}_C[|\Pr(Z = 1 | C) - C|]$ . In practice, after partitioning  $N$  examples into  $B$  confidence bins  $\{\mathcal{I}_b\}_{b=1}^B$ , we estimate ECE as

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{I}_b|}{N} |\text{acc}(\mathcal{I}_b) - \text{conf}(\mathcal{I}_b)|. \quad (4)$$

Here,  $\text{acc}(\mathcal{I}_b)$  and  $\text{conf}(\mathcal{I}_b)$  are the mean correctness and confidence in bin  $b$ , respectively. While ECE measures calibration at the bin level, the **Brier score** (Brier, 1950) measures the squared error between confidence and correctness for each instance,

$$\text{Brier} = \frac{1}{N} \sum_{j=1}^N (c_j - z_j)^2, \quad (5)$$

where lower values indicate better probabilistic predictions for both metrics. These metrics assess calibration, but neither directly evaluates discrimination as a ranking property. For example, assigning every response a constant confidence equal to the overall accuracy yields zero ECE while giving correct and incorrect responses identical scores (Eusebi et al., 2026). We therefore measure the discrimination using **AUROC**, which evaluates whether correct responses receive higher confidence than incorrect ones.

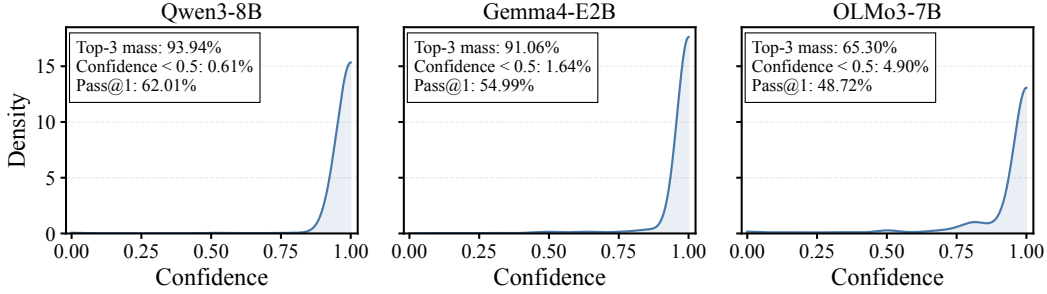


Figure 1: Confidence distributions of three representative open-source LLMs on DeepScaleR. All three models concentrate probability mass near 1. Fewer than 5% of outputs have confidence below 0.5 for each model.

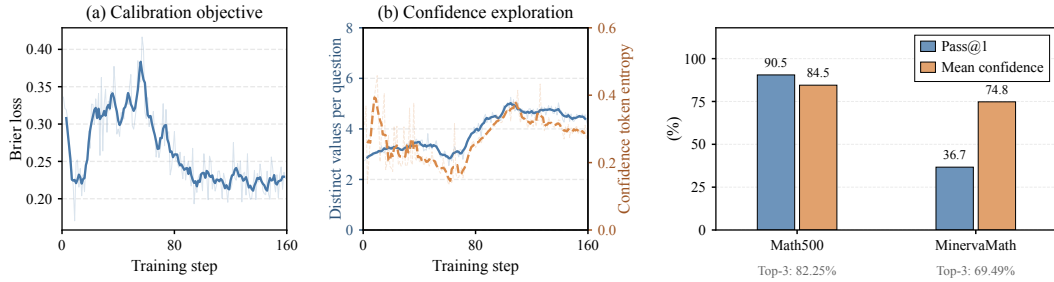


Figure 2: Training dynamics of Qwen3-8B with RLCR. (a) Brier loss. (b) Distinct confidence values per question (eight rollouts; blue, left axis) and confidence token entropy (orange, right axis). Bold curves show centered five-step moving averages; faint curves show per-step measurements.

Figure 3: Test-time calibration at step 160 on Math500 and MinervaMath (ID), using four rollouts per question. Bars show Pass@1 and mean confidence; task labels report Top-3 mass.

### 3 MOTIVATION

Confidence-aware RL typically optimizes calibration rewards on responses sampled from LLMs (Damani et al., 2026). To investigate whether LLMs can learn calibrated confidence estimates through RL, we analyze how their confidence distributions change during training.

**Confidence priors concentrate on a few high values.** We refer to the confidence distribution before RL as the model’s *confidence prior*. To characterize this prior, we sample 10 rollouts per question from Qwen3-8B (Yang et al., 2025), Gemma4-E2B-Instruct (Abd et al., 2026), OLMo3-7B-Instruct (Ettinger et al., 2025) on DeepScaleR-Train, and present their confidence distributions in Figure 1. All three models concentrate their confidence on a few high values, with fewer than 5% of outputs below 0.5. These base models also assign high confidence even to incorrect responses, suggesting that their verbalized confidence does not reliably reflect answer correctness. As a result, RL starts from a sub-optimal confidence prior that rarely produces low-confidence samples.

**RL improves calibration but confidence exploration remains limited.** To examine whether calibration-aware RL overcomes this concentration, we train Qwen3-8B with RLCR. As shown in Figure 2(a), RLCR successfully reduces Brier loss during training, demonstrating its effectiveness in improving calibration. However, Figure 2(b) shows that the model continues to sample a small set of confidence values for each question. We provide more detailed empirical evidence in Appendix B.2.

To better understand why this concentration persists during training, we formulate confidence generation as action selection under a softmax policy and examine how sampling probabilities affect the expected gradient (Garg et al., 2022).

**Proposition 1** (Policy gradients under confidence concentration). *Fix prediction information  $h$  and a finite set of confidence actions  $\{c_1, \dots, c_K\} \subseteq [0, 1]$ . Let  $\pi_i = \Pr(C = c_i \mid h) = \exp(\ell_i) / \sum_{j=1}^K \exp(\ell_j)$ . For  $Z \in \{0, 1\}$  indicating answer correctness, define the fixed expected reward  $r_i = -\mathbb{E}[(c_i - Z)^2 \mid h]$ . Let  $J_h(\ell) = \sum_i \pi_i r_i$  and  $\Delta_h = \max_i r_i - \min_i r_i$ . Then*

$$\frac{\partial J_h}{\partial \ell_i} = \pi_i(r_i - J_h), \quad \left| \frac{\partial J_h}{\partial \ell_i} \right| \leq \pi_i \Delta_h. \quad (6)$$

*If  $\pi_k = 1 - \varepsilon$  for some  $k \in \{1, \dots, K\}$ , then*

$$\|\nabla_{\ell} J_h\|_1 \leq 2\varepsilon \Delta_h. \quad (7)$$

Proposition 1 shows that a rarely sampled confidence value has a small expected logit gradient even when choosing it would improve the expected reward. As the confidence distribution approaches a single choice, the entire logit gradient vanishes, including when that choice is suboptimal. This provides a mechanism by which an initially concentrated confidence distribution can impede exploration of better-calibrated outputs. We provide the proof in Appendix B.3.

**Persistent confidence concentration hinders downstream calibration.** To examine the implications of persistent confidence concentration for downstream calibration, we evaluate the final RLCR checkpoint on two datasets of contrasting difficulty including Math500 and MinervaMath. As shown in Figure 3, the post-RL model underestimates its accuracy on Math500 but is markedly overconfident on MinervaMath. Despite the large accuracy gap, the mean confidence remains similar across the two tasks, with the three most frequent values accounting for about 70% of outputs on each tasks. This suggests that confidence estimates do not sufficiently adapt to differences in task difficulty, and further contribute to downstream miscalibration.

To characterize the calibration cost of this mismatch, we examine the expected Brier loss, which is minimized when confidence equals the conditional probability of correctness (Gneiting & Raftery, 2007; Damani et al., 2026). The following proposition quantifies how little probability mass near this optimal confidence raises the lower bound on Brier loss.

**Proposition 2** (Brier risk under limited confidence sampling). *Let  $Z \in \{0, 1\}$  indicate answer correctness and let  $C \in [0, 1]$  denote the predicted confidence. Given prediction information  $h$ , define  $p_h = \Pr(Z = 1 \mid h)$ . Assume that  $C$  and  $Z$  are conditionally independent given  $h$ . For any  $\delta > 0$ , let  $m_\delta(h) = \Pr(|C - p_h| < \delta \mid h)$ . Then*

$$\mathbb{E}[(C - Z)^2 \mid h] \geq p_h(1 - p_h) + \delta^2(1 - m_\delta(h)). \quad (8)$$

For a fixed neighborhood of  $p_h$ , Proposition 2 shows that the lower bound on Brier loss increases as the probability of sampling confidence within that neighborhood decreases. This suggests that confidence concentration can hinder calibration when it leaves little probability mass near the probability of correctness. We provide the proof, the restricted-support case, and a finite-sampling analysis in Appendix B.4.

## 4 METHOD

To mitigate these limitations, we introduce CalibSFT, an offline supervised initialization for post-answer verbalized confidence with two components. It constructs rollout-derived confidence targets with a balanced target distribution, then applies outcome-conditional segment supervision. Compared with the Base initialization, it changes only this offline initialization; the downstream confidence-RL objective and training recipe are held fixed.

### 4.1 ROLLOUT-DERIVED CONFIDENCE TARGETS

For each training prompt  $x_g$ , where  $g$  indexes prompt groups, we sample  $K_{\text{SFT}}$  completions indexed by  $i$  and obtain binary correctness labels  $z_{g,i}$  from the task verifier. We compute the group success rate from all sampled completions before filtering, and assign a target to each retained rollout:

$$q_g = \frac{1}{K_{\text{SFT}}} \sum_{i=1}^{K_{\text{SFT}}} z_{g,i}, \quad c_{g,i}^* = \lambda q_g + (1 - \lambda) z_{g,i}. \quad (9)$$

The group statistic is the empirical success rate under the initial policy and decoding protocol, whereas  $z_{g,i}$  records the outcome of the realized answer. Thus  $\lambda$  controls the interpolation between prompt-level and instance-level supervision; we use  $\lambda = 0.5$  in the main configuration. Invalid or low-quality completions are removed before they enter the SFT set; this filtering does not recompute  $q_g$ . Replacing the confidence suffix with  $c_{g,i}^*$  yields a target-rendered completion that retains the sampled reasoning and answer. In the main probability format, the target is separately rendered as a two-decimal value in the trailing `<confidence>` tag.

We balance the target distribution at training time: a per-epoch sampler groups examples using a six-decimal key of the numeric target and samples the same number from each nonempty bucket, using the smallest bucket as the default per-target count. For runs using offline materialization, the same rule is applied when constructing the balanced split. This exposes the model to the available confidence targets without inheriting their raw rollout-frequency skew.

## 4.2 OUTCOME-CONDITIONAL SUPERVISION

For a target-rendered completion  $\tilde{o}$ , let  $M$  denote its reasoning-and-answer token positions and  $C$  its trailing confidence positions. CalibSFT uses token-mean cross-entropy with separate averages for the two segments:

$$\begin{aligned} \mathcal{L}_{\text{CalibSFT}} = (1 - \beta) \mathbb{E}_{\text{correct}} & \left[ -\frac{1}{|M|} \sum_{t \in M} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right] \\ & + \beta \mathbb{E}_{\text{all}} \left[ -\frac{1}{|C|} \sum_{t \in C} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right]. \end{aligned} \quad (10)$$

The first expectation is over retained correct rollouts and the second over all retained rollouts. A correct rollout therefore trains both its reasoning/answer and confidence segments, whereas an incorrect rollout trains only its confidence suffix. We use  $\beta = 0.5$  in the main configuration.

## 5 EXPERIMENTS

### 5.1 EXPERIMENTAL SETUP

**Baselines.** We compare methods spanning answer-only and calibration-aware training. **Base** denotes the checkpoint without task-specific training, while **RLVR** optimizes only answer correctness. We use confidence-aware RL baselines including **RLCR** (Damani et al., 2026), **ReDoubt** (Bani-Harouni et al., 2026), **CoCA** (Li et al., 2026), **DCPO** (Ma et al., 2026), and **C3RL** (Yang et al., 2026). To measure the effect of **CalibSFT**, we evaluate each confidence-aware RL method from both the Base and CalibSFT checkpoints. All methods use Qwen3-8B (Yang et al., 2025) as the default model and follow the **DAPO** training recipe (Yu et al., 2025).

**Datasets.** We use **DeepScaleR-Preview** (Luo et al., 2025) as the mathematical training dataset. We evaluate on a broad range of in-distribution (ID) and out-of-domain (OOD) datasets spanning different levels of difficulty. For ID mathematical reasoning, we use **DeepScaleR-Eval**, **MATH-500** (Hendrycks et al., 2021), **MinervaMath** (Lewkowycz et al., 2022), **OlympiadBench** (He et al., 2024), **GSM8K** (Cobbe et al., 2021), and **AIME 2024–2026** (Dekoninck et al., 2026). For general reasoning, we use **HotpotQA** (Yang et al., 2018), **MuSiQue** (Trivedi et al., 2022), **TriviaQA** (Joshi et al., 2017), **NQOpen** (Lee et al., 2019), **PopQA** (Mallen et al., 2023), **WebQuestions** (Berant et al., 2013), **DROP** (Dua et al., 2019), and **LiveBenchReasoning** (White et al., 2024).

**Evaluation metrics.** We evaluate model performance using complementary metrics that capture answer correctness, confidence discrimination, and absolute calibration. We report **Pass@1** for answer correctness, **AUROC** for ranking correct responses above incorrect responses, and **Brier score** (Brier, 1950) and **ECE** (Guo et al., 2017) for probabilistic calibration.

**Implementation details.** During evaluation, we use temperature 0.6 and sample 32 rollouts per problem for the AIME benchmarks and 4 rollouts per problem for all other benchmarks. All

Table 1: Average performance across 8 ID and 8 OOD datasets. Higher is better for Pass@1 and AUROC; lower is better for Brier and ECE. Each entry averages the datasets within the corresponding split. Base, RLVR, and standalone CalibSFT use a matched 6,144-token evaluation budget.

| Method     | In-distribution   |                  |                    |                  | Out-of-distribution |                  |                    |                  |
|------------|-------------------|------------------|--------------------|------------------|---------------------|------------------|--------------------|------------------|
|            | Pass@1 $\uparrow$ | AUROC $\uparrow$ | Brier $\downarrow$ | ECE $\downarrow$ | Pass@1 $\uparrow$   | AUROC $\uparrow$ | Brier $\downarrow$ | ECE $\downarrow$ |
| Base       | 57.39             | 71.47            | 34.97              | 36.26            | 45.36               | <b>65.71</b>     | 45.15              | 46.94            |
| RLVR       | <b>61.60</b>      | 73.16            | 33.48              | 34.08            | 43.88               | 64.19            | 50.29              | 51.52            |
| CalibSFT   | 56.20             | <b>79.25</b>     | <b>18.01</b>       | <b>14.82</b>     | <b>45.91</b>        | 61.38            | <b>28.25</b>       | <b>24.16</b>     |
| RLCR       | 62.71             | 82.78            | 15.60              | 16.84            | 43.96               | 69.38            | 31.14              | 31.17            |
| + CalibSFT | <b>64.70</b>      | <b>86.88</b>     | <b>13.48</b>       | <b>10.84</b>     | <b>45.71</b>        | <b>71.31</b>     | <b>23.24</b>       | <b>18.20</b>     |
| ReDoubt    | –                 | –                | –                  | –                | –                   | –                | –                  | –                |
| + CalibSFT | –                 | –                | –                  | –                | –                   | –                | –                  | –                |
| CoCA       | 61.11             | 80.59            | 17.25              | 21.75            | 43.68               | 66.79            | 26.67              | 24.12            |
| + CalibSFT | <b>62.65</b>      | <b>86.31</b>     | <b>16.40</b>       | <b>20.21</b>     | <b>45.55</b>        | <b>71.21</b>     | <b>22.95</b>       | <b>18.89</b>     |
| DCPO       | 61.98             | 78.16            | 16.38              | 14.68            | 43.99               | 68.13            | 32.85              | 32.41            |
| + CalibSFT | <b>62.40</b>      | <b>85.16</b>     | <b>13.29</b>       | <b>12.20</b>     | <b>44.82</b>        | <b>72.75</b>     | <b>21.73</b>       | <b>17.31</b>     |
| C3RL       | 63.22             | 67.90            | 27.80              | 30.45            | 43.44               | 62.19            | 45.54              | 48.00            |
| + CalibSFT | <b>63.32</b>      | <b>80.88</b>     | <b>15.65</b>       | <b>15.18</b>     | <b>44.60</b>        | <b>63.03</b>     | <b>23.89</b>       | <b>17.38</b>     |

experiments are conducted on a single node with eight NVIDIA A100 GPUs. We report our hyperparameter setting and training details in Appendix C.

## 5.2 OVERALL RESULTS

**CalibSFT improves calibration across diverse confidence-aware RL methods.** As shown in Table 1, initializing RL with CalibSFT reduces Brier score and ECE on mathematical reasoning benchmarks for all three completed paired comparisons: RLCR, CoCA, and DCPO. AUROC also improves for all three methods. We provide detailed per-dataset results in Table 7 (Appendix D.1). For example, CalibSFT reduces DCPO’s ECE from 44.59 to 20.25 while increasing its AUROC from 64.19 to 74.73 on MinervaMath. CalibSFT achieves these gains while largely preserving answer accuracy, with an average Pass@1 gain of 1.32 points across these three RL methods. These results support its use as a plug-and-play initialization for better confidence-aware RL.

**CalibSFT generalizes to general reasoning tasks beyond the training domain.** To evaluate the generalization of CalibSFT beyond its mathematical domain, we assess the resulting models on eight general reasoning benchmarks. Table 1 shows lower average Brier score and ECE for all three completed paired comparisons on these benchmarks. For example, CalibSFT reduces DCPO’s ECE from 20.53 to 8.55 on MuSiQue. We provide detailed per-dataset results in Table 8 (Appendix D.1).

**CalibSFT improves confidence–accuracy correlation across tasks of varying difficulty.** To assess whether average confidence tracks accuracy across tasks, we compare each dataset’s average confidence with its accuracy. For each RL method with and without CalibSFT initialization, we compute Pearson and Spearman correlations separately over the eight ID and eight OOD datasets. Table 2 shows that CalibSFT improves Spearman correlation for all three completed RL comparisons on mathematical and general reasoning benchmarks. For example, DCPO’s Spearman correlation increases from 0.619 to 0.976 on mathematical tasks and from 0.548 to 0.643 on general reasoning tasks. Pearson correlation also improves for DCPO and for RLCR on ID tasks, while remaining essentially unchanged for RLCR on OOD tasks. These results suggest that CalibSFT helps downstream RL models assign higher average confidence to tasks on which they perform better.

**CalibSFT mitigates confidence concentration after downstream RL.** Prior work shows that confidence estimates can remain concentrated on a few values even after calibration-aware RL training (Wang et al., 2026; Yang et al., 2026). To examine how CalibSFT initialization influences confidence distribution, we compare RLCR, CoCA, and DCPO on MinervaMath. As shown in Figure 4, models trained without CalibSFT tend to concentrate their confidence on a few values

Table 2: Correlation between accuracy and mean verbalized confidence across ID and OOD datasets. Base and +Ours denote RL without and with CalibSFT initialization, respectively; the higher value is bold. Each coefficient uses eight equally weighted datasets within its split.

| Dataset | Metric          | RLCR  |              | ReDoubt |       | CoCA  |              | DCPO  |              | C3RL  |              |
|---------|-----------------|-------|--------------|---------|-------|-------|--------------|-------|--------------|-------|--------------|
|         |                 | Base  | +Ours        | Base    | +Ours | Base  | +Ours        | Base  | +Ours        | Base  | +Ours        |
| ID      | Pearson $r$     | 0.768 | <b>0.963</b> | –       | –     | 0.750 | <b>0.887</b> | 0.669 | <b>0.996</b> | 0.635 | <b>0.896</b> |
|         | Spearman $\rho$ | 0.619 | <b>0.952</b> | –       | –     | 0.643 | <b>0.976</b> | 0.619 | <b>0.976</b> | 0.595 | <b>0.833</b> |
| OOD     | Pearson $r$     | 0.505 | <b>0.729</b> | –       | –     | 0.447 | <b>0.605</b> | 0.467 | <b>0.611</b> | 0.681 | <b>0.848</b> |
|         | Spearman $\rho$ | 0.548 | <b>0.619</b> | –       | –     | 0.571 | <b>0.619</b> | 0.548 | <b>0.643</b> | 0.714 | <b>0.929</b> |

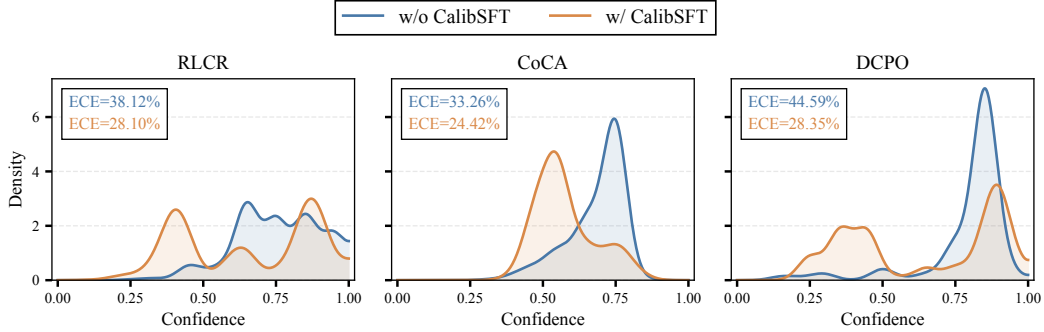


Figure 4: Confidence distributions on MinervaMath for RLCR, CoCA, and DCPO, with four rollouts per question at temperature 0.6. CalibSFT reduces concentration at dominant confidence values and lowers ECE for all three methods.

and show higher average confidence than their accuracy. CalibSFT reduces the dominant peaks and brings average confidence closer to accuracy, alleviating both concentration and overconfidence. For example, DCPO’s average confidence decreases from 79.32% to 63.69%, while accuracy increases from 34.93% to 36.58%. ECE decreases for all three methods, from 44.59% to 28.35% for DCPO. We provide additional confidence distribution metrics in Appendix D.7.

**The calibration gains of CalibSFT persist across reasoning budgets.** To assess whether CalibSFT improves calibration as reasoning progresses rather than the final answer alone, we compare RLCR with and without CalibSFT initialization on AIME 2026 under varying token budgets. As shown in Figure 5, CalibSFT yields substantially lower ECE across all token budgets. It matches the RLCR baseline in accuracy around 5,000 tokens and becomes more accurate at larger budgets. For example, CalibSFT reduces ECE from 22.71 to 10.97 at a 10,000-token budget, while accuracy increases from 47.50% to 52.71%. These results suggest that CalibSFT’s calibration benefits persist as more computation is allocated to reasoning.

### 5.3 ABLATION STUDIES

**Confidence target construction.** The construction of confidence targets in CalibSFT is inspired by proper scoring rules, which serve as a foundation in calibration. To verify the effectiveness of this target design, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024), semantic consistency (SaySelf) (Xu et al., 2024), and rubric-based LLM judgments (LoVeC) (Zhang et al., 2026), using a shared SFT training set the same optimization objective. We provide details of the subset and target construction in Appendix D.2. We report the results of the target construction in Table 3, which shows the AUROC and Brier score of the constructed targets before SFT and average performance on ID mathematical benchmarks after SFT. CalibSFT yields lower Brier score and higher AUROC than the four alternatives. These results

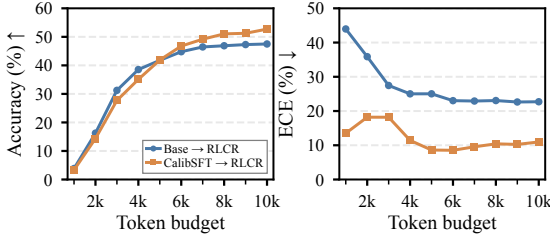


Figure 5: Accuracy (left) and ECE (right) on AIME’26 across token budgets for RLCR with and without CalibSFT initialization. CalibSFT yields substantially lower ECE across all budgets while matching accuracy at medium budgets and improving it at larger budgets.

| Target   | Pre-SFT targets  |                    | Post-SFT predictions |                    |
|----------|------------------|--------------------|----------------------|--------------------|
|          | AUROC $\uparrow$ | Brier $\downarrow$ | AUROC $\uparrow$     | Brier $\downarrow$ |
| CSFT     | 50.00            | 31.67              | 73.46                | 19.15              |
| UaIT     | 52.05            | 33.17              | 74.72                | 24.24              |
| SaySelf  | –                | 41.33              | 57.69                | 39.38              |
| LoVeC    | 75.40            | 24.71              | 71.11                | 32.32              |
| CalibSFT | <b>100.00</b>    | <b>7.92</b>        | <b>79.75</b>         | <b>17.81</b>       |

Table 3: The calibration comparison between pre-SFT constructed targets and post-SFT predictions. CalibSFT achieves higher AUROC and lower Brier score across ID mathematical benchmarks.

Table 5: Supervision settings and average performance on eight ID benchmarks. A check mark indicates that a segment contributes to the training loss; segment balancing averages the reasoning/answer and confidence losses separately. All variants use the same statically balanced training set and train for 180 steps.

| Variant         | Reasoning/answer |           | Confidence |           | Segment balanced | ID performance    |                  |                    |                  |
|-----------------|------------------|-----------|------------|-----------|------------------|-------------------|------------------|--------------------|------------------|
|                 | Correct          | Incorrect | Correct    | Incorrect |                  | Pass@1 $\uparrow$ | AUROC $\uparrow$ | Brier $\downarrow$ | ECE $\downarrow$ |
| Confidence-only | –                | –         | ✓          | ✓         | –                | 43.56             | 74.62            | 20.97              | 17.43            |
| Full-CE         | ✓                | ✓         | ✓          | ✓         | –                | 65.89             | 61.66            | 29.29              | 27.65            |
| Full-Balanced   | ✓                | ✓         | ✓          | ✓         | ✓                | 65.46             | 80.67            | 17.04              | <b>13.87</b>     |
| Correct-only    | ✓                | –         | ✓          | –         | ✓                | <b>66.59</b>      | <b>81.58</b>     | <b>16.95</b>       | 14.28            |
| CalibSFT        | ✓                | –         | ✓          | ✓         | ✓                | 65.79             | 79.30            | 17.89              | 14.36            |

support the effectiveness of our confidence target construction, which accounts for both calibration and discrimination.

**Target mixing weight.** We construct confidence targets as  $c_{g,i}^* = \lambda q_g + (1 - \lambda) z_{g,i}$ , where  $q_g$  is the group success rate and  $z_{g,i}$  is the correctness of an individual response. To evaluate the effect of the mixing weight, we compare  $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$  using the same subset as above and keeping the SFT settings fixed. At  $\lambda = 0$ , targets are restricted to 0 and 1; this setting yields the highest Brier score and ECE in Table 4. At  $\lambda = 1$ , correct and incorrect responses to the same question receive identical targets, providing no within-question discrimination signal; post-SFT AUROC is also lower than at  $\lambda = 0.5$ . Equal weighting achieves the highest AUROC and lowest Brier score and ECE among the tested settings, so we use  $\lambda = 0.5$  as the default.

Table 4: Average calibration performance on ID benchmarks across mixing weights. Equal weighting ( $\lambda = 0.5$ ) performs best on all three metrics.

| $\lambda$  | AUROC $\uparrow$ | Brier $\downarrow$ | ECE $\downarrow$ |
|------------|------------------|--------------------|------------------|
| 0          | 68.10            | 32.49              | 32.49            |
| 0.25       | 79.02            | 18.83              | 14.21            |
| <b>0.5</b> | <b>79.75</b>     | <b>17.81</b>       | <b>13.03</b>     |
| 0.75       | 65.93            | 19.99              | 13.16            |
| 1          | 73.29            | 19.90              | 15.04            |

**Supervision design.** [TODO:] Compare confidence-only SFT, full-completion CE, segment-balanced full-completion CE, CalibSFT, and CalibSFT without target-balanced sampling. Hold the training examples, confidence targets, and optimization settings fixed across the supervision variants. Report which completion segments contribute to each objective, followed by their ID/OOD Pass@1 and calibration results. Use this ablation to isolate the effects of retaining answer supervision on correct rollouts, masking incorrect reasoning and answers, supervising confidence on all rollouts, and balancing confidence targets.

## 6 DISCUSSION

### 6.1 VERBALIZED CONFIDENCE AND ALTERNATIVE CONFIDENCE SIGNALS

**Verbalized confidence provides a directly interpretable uncertainty signal.** [TODO:] Compare post-answer verbalized confidence with token- or sequence-level probability, self-consistency, semantic consistency, and a hidden-state probe. Use the same generated answers whenever the signal permits, and report AUROC, Brier score, ECE, inference cost, and model-access requirements. Discuss which gains arise from CalibSFT and which arise from the choice of confidence signal.

### 6.2 CONFIDENCE FORMATS AND RESOLUTION

**CalibSFT can be instantiated with different confidence formats.** [TODO:] Compare two-decimal probabilities, discrete confidence levels, and semantic labels such as *low*, *medium*, and *high*. Map all formats to  $[0, 1]$  for evaluation and report Pass@1, AUROC, Brier score, and ECE. Include ConfTuner as the representative tokenized-confidence method when its output format and evaluation mapping are matched.

**The rollout count controls the resolution of empirical targets.** [TODO:] Fix  $\lambda = 0.5$  and compare target construction with  $K_{\text{SFT}} \in \{10, 20, 100\}$ . Report the resulting target coverage, target estimation cost, and final calibration. Use this comparison to separate the effect of a finer confidence grid from the effect of additional rollout samples.

## 7 DOWNSTREAM APPLICATIONS

### 7.1 SELECTIVE ANSWERING AND RISK CONTROL

[TODO:] Rank responses by confidence and report risk–coverage curves, AURC, and accuracy at matched coverage. Compare Base, confidence-aware RL, and CalibSFT-initialized RL to test whether the calibration improvements translate into more reliable abstention decisions.

### 7.2 CONFIDENCE-GUIDED EFFICIENT REASONING

[TODO:] Use confidence to allocate additional generation budget or route uncertain questions to a stronger inference procedure. Compare against fixed-budget and random-routing baselines at matched token cost, reporting accuracy, calibration, and total generated tokens.

## 8 CONCLUSION

[TODO:] Summarize the concentrated and misaligned confidence prior, the rollout-derived and target-balanced CalibSFT initialization, and its effect on downstream confidence-aware RL. Close with the main ID/OOD evidence and the remaining limitations of rollout-derived confidence targets.

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## AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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## A RELATED WORK

### A.1 CONFIDENCE ESTIMATION IN LLMs

LLM confidence can be estimated from token probabilities, hidden-state probes, agreement across sampled answers, or explicit self-reports (Kadavath et al., 2022; Azaria & Mitchell, 2023; Kuhn et al., 2023; Xiong et al., 2024). Verbalized-confidence methods ask the model to emit a numerical or linguistic confidence score, providing a single-generation estimate without requiring access to internal model representations (Tian et al., 2023; Xiong et al., 2024). Our work studies this post-answer verbalized setting.

## A.2 CONFIDENCE-AWARE POST-TRAINING

Supervised approaches train confidence outputs using correctness labels, empirical rollout frequencies, or confidence-token objectives (Lin et al., 2022; Xu et al., 2024; Li et al., 2025). Calibration-aware RL instead incorporates proper scoring rewards or modifies the confidence target and credit assignment. Representative methods include RLCR, Rewarding Doubt, CoCA, DCPO, and CALIBER (Damani et al., 2026; Bani-Harouni et al., 2026; Li et al., 2026; Ma et al., 2026; Finlay et al., 2026). CalibSFT focuses on the supervised initialization supplied to these downstream RL objectives.

## B ADDITIONAL ANALYSIS OF THE MOTIVATION

### B.1 COMPLETE CONFIDENCE DISTRIBUTIONS

For the initial confidence analysis in Section 3, each model generates ten responses to each of the same 36,915 DeepScaleR-Train questions, with temperature 1.0, top- $p$  0.95, and an 8,192-token budget. We analyze the original confidence outputs before target construction. Valid-confidence rates are 80.57%, 99.81%, and 79.49% for Qwen3-8B, Gemma4-E2B-Instruct, and OLMo3-7B-Instruct, respectively. Distribution statistics and correctness-conditional confidence means use valid outputs. Figure 1 uses Gaussian smoothing with bandwidth 0.04 and reflection at zero and one; Confidence values are rounded to two decimal places before smoothing; frequency statistics use the rounded, unsmoothed values.

### B.2 CONFIDENCE SAMPLING DURING RL

Table 6 reports Base→RLCR statistics averaged over training steps. Confidence token entropy is the mean next-token entropy at confidence-value token positions. Top-3 share is the fraction of a step’s sampled confidence values assigned to its three most frequent values. Distinct confidence counts are computed within each question’s group of eight responses and then averaged. Statistics use the groups retained by the training filter; the sampled questions and responses can change during training. The logged confidence values include zero placeholders for missing outputs. Excluding all zero values, the final 40 steps still have a mean top-3 share of 45.16%.

Table 6: Brier loss and confidence sampling during Base→RLCR training. Later training lowers Brier loss while sampled confidence values remain concentrated.

| Steps   | Brier | Confidence token entropy | Top-3 share (%) | Distinct values per group |
|---------|-------|--------------------------|-----------------|---------------------------|
| 1–10    | 0.266 | 0.325                    | 81.00           | 2.95                      |
| 11–40   | 0.293 | 0.244                    | 76.51           | 3.26                      |
| 41–80   | 0.310 | 0.196                    | 74.15           | 3.26                      |
| 81–120  | 0.230 | 0.322                    | 46.07           | 4.58                      |
| 121–160 | 0.225 | 0.308                    | 44.04           | 4.59                      |

### B.3 PROOF OF PROPOSITION 1

*Proof.* We fix  $h$  and the expected rewards  $r_i$ , and differentiate with respect to the confidence logits. The softmax derivative is  $\partial\pi_j/\partial\ell_i = \pi_j(\mathbf{1}\{j = i\} - \pi_i)$ . It follows that

$$\frac{\partial J_h}{\partial\ell_i} = \sum_j r_j \pi_j (\mathbf{1}\{j = i\} - \pi_i) \quad (11)$$

$$= \pi_i (r_i - J_h). \quad (12)$$

Since  $J_h$  is a convex combination of the  $r_i$  values,  $|r_i - J_h| \leq \Delta_h$ , which gives the coordinate-wise bound. If  $\pi_k = 1 - \varepsilon$ , then

$$|r_k - J_h| = \left| \sum_{i \neq k} \pi_i (r_k - r_i) \right| \leq \varepsilon \Delta_h. \quad (13)$$

Consequently,

$$\|\nabla_\ell J_h\|_1 = \pi_k |r_k - J_h| + \sum_{i \neq k} \pi_i |r_i - J_h| \quad (14)$$

$$\leq (1 - \varepsilon)\varepsilon\Delta_h + \varepsilon\Delta_h \quad (15)$$

$$\leq 2\varepsilon\Delta_h. \quad (16)$$

□

The gradient is the expectation of the likelihood-ratio estimator  $R\nabla_\ell \log \pi_I$ , where  $I \sim \pi$  and  $R = -(c_I - Z)^2$ , with  $Z$  drawn from its fixed conditional distribution given  $h$ , independently of  $I$ . A baseline depending only on  $h$  leaves this expectation unchanged.

#### B.4 PROOF OF PROPOSITION 2

*Proof.* Conditional independence gives  $\mathbb{E}[Z - p_h \mid h, C] = 0$ . Expanding the square therefore yields

$$\mathbb{E}[(C - Z)^2 \mid h] = \mathbb{E}[(Z - p_h)^2 \mid h] + \mathbb{E}[(C - p_h)^2 \mid h] \quad (17)$$

$$= p_h(1 - p_h) + \mathbb{E}[(C - p_h)^2 \mid h]. \quad (18)$$

For any  $\delta > 0$ , the second term satisfies

$$\mathbb{E}[(C - Z)^2 \mid h] - p_h(1 - p_h) = \mathbb{E}[(C - p_h)^2 \mid h] \quad (19)$$

$$\geq \delta^2 \Pr(|C - p_h| \geq \delta \mid h) \quad (20)$$

$$= \delta^2(1 - m_\delta(h)). \quad (21)$$

Thus, assigning little probability mass near  $p_h$  incurs excess Brier loss even when confidence values in that neighborhood remain possible. □

**Restricted confidence support.** As a special case, suppose  $\Pr(C \in S_h \mid h) = 1$  for a nonempty set  $S_h \subseteq [0, 1]$ , and define  $d_h = \text{dist}(p_h, S_h) = \inf_{c \in S_h} |c - p_h|$ . If  $d_h > 0$ , then  $m_{d_h}(h) = 0$ , and Proposition 2 with  $\delta = d_h$  gives

$$\mathbb{E}[(C - Z)^2 \mid h] \geq p_h(1 - p_h) + \text{dist}(p_h, S_h)^2. \quad (22)$$

For  $d_h = 0$ , the same bound follows directly from the decomposition above. If  $S_h$  is closed, a predictor that always outputs a nearest point in  $S_h$  attains the bound.

**Finite-sampling limitation.** Fix the policy and prediction information  $h$ , and draw  $G$  independent confidence samples  $C_1, \dots, C_G$  from its conditional distribution. Each confidence sample lies outside  $\{c : |c - p_h| < \delta\}$  with probability  $1 - m_\delta(h)$ . Independence gives the probability  $(1 - m_\delta(h))^G$  that all  $G$  samples lie outside. On that event, even the sample closest to  $p_h$  has squared error at least  $\delta^2$ , so

$$\mathbb{E}\left[\min_{i \leq G} (C_i - p_h)^2 \mid h\right] \geq \delta^2(1 - m_\delta(h))^G. \quad (23)$$

Here  $p_h$  is conditional on the information used for a particular prediction, rather than the prompt’s average success rate across different answers. The sampling statement fixes this information and the policy; it does not assume that support remains unchanged during RL.

#### B.5 ADDITIONAL MODELS AND SAMPLING SETTINGS

**[TODO:]** Add the motivation analysis for additional model families and sampling settings once those evaluation runs are complete.

## C IMPLEMENTATION DETAILS

### C.1 TRAINING PIPELINES

All runs use Qwen3-8B as the base model. We randomly hold out 2,000 examples from DeepScaleR-Preview as a test split (DeepScaleR-Eval) and use all remaining examples as the training split (DeepScaleR-Train), which is used to construct the SFT data for CalibSFT and the downstream RL training data.

**CalibSFT.** To construct CalibSFT targets, we sample 10 rollouts per prompt from DeepScaleR-Train with temperature 1.0 and top- $p$  0.95. We compute its group success rate, and combine group-level and rollout-level correctness with  $\lambda = 0.5$ . We use target-balanced sampling and the selective completion mask with  $\beta = 0.5$ : correct rollouts supervise both the reasoning/answer and confidence segments, whereas incorrect rollouts supervise only the confidence segment. Generally, we train CalibSFT for 180 steps with a learning rate of  $2 \times 10^{-6}$  and 10 warm-up steps.

**Confidence-aware RL.** For downstream RL, each prompt group contains 8 sampled responses, with each response capped at 8,192 tokens. We train with batches of 64 prompt groups for 160 optimization steps. For methods using DAPO dynamic sampling, this corresponds to approximately 600 rollout-generation batches. Dynamic sampling retains only groups with mixed correctness outcomes and discards groups whose responses are all correct or all incorrect. To accelerate training for these methods, we remove questions for which all sampled responses from the base model are correct before RL training begins. Each method retains its own reward and credit assignment.

- **Direct RL baselines.** RLVR, RLCR (Damani et al., 2026), CoCA (Li et al., 2026), and DCPO (Ma et al., 2026) are trained directly from the Qwen3-8B base checkpoint. They share the DAPO training recipe but retain their respective correctness, calibration-target, and credit-assignment objectives.
- **ReDoubt.** ReDoubt is initialized from the trained RLVR checkpoint and then optimized with its clipped log-score confidence objective (Bani-Harouni et al., 2026). When incorporated with CalibSFT, we first train CalibSFT on the RLVR checkpoint and then continue with ReDoubt RL.
- **C3RL.** C3RL (Yang et al., 2026) combines correctness, calibration, and reference-accuracy rewards. C3RL uses the digit confidence format, expressing confidence as an integer from 0 to 9 and mapping it to  $[0, 1]$  by dividing by 9. The CalibSFT initialization for C3RL is also trained in this format. We use the base model’s rollout success rates to label each training question as all correct, partially correct, or all incorrect. The reference reward encourages correct answers to previously all-incorrect questions and penalizes incorrect answers to previously all-correct questions. To retain these questions and their reference-reward signals, we disable DAPO dynamic sampling and all-correct filtering, using the full RL training split for C3RL both with and without CalibSFT initialization.

## C.2 EVALUATION PROTOCOL

For the main results, we use temperature 0.6 and sample four rollouts per question on non-AIME mathematical and OOD benchmarks and 32 rollouts per question on AIME. Pass@1 is the mean correctness over these sampled rollouts. The supervision and target ablations retain their recorded evaluation settings. Confidence is parsed from the final probability field and normalized to  $[0, 1]$ . We use 10 equal-width bins for calculating ECE. Unless otherwise noted, RL results use the final checkpoint after 160 optimization steps, while SFT-only results use the checkpoint at the end of their respective SFT runs.

## D DETAILED EXPERIMENTAL RESULTS

### D.1 PER-DATASET MAIN RESULTS

Table 1 reports split-level averages. The following tables expose the same metrics for every benchmark; the final cell in each split is the unweighted average over its datasets.

### D.2 ABLATION OF SFT CONFIDENCE TARGETS

To evaluate the advantage of confidence targets derived from a proper scoring rule, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024), semantic consistency (SaySelf) (Xu et al., 2024), and

Table 9: Training-target distribution (%) over ten equal-width bins in the confidence target ablation.

|    | Interval   | CSFT  | UaIT | SaySelf | LoVeC | CalibSFT |
|----|------------|-------|------|---------|-------|----------|
|    | [0.0, 0.1] | 11.11 | 0.00 | 22.07   | 27.33 | 11.11    |
|    | (0.1, 0.2] | 11.11 | 0.00 | 14.41   | 3.78  | 11.11    |
|    | (0.2, 0.3] | 11.11 | 0.00 | 11.93   | 8.17  | 11.11    |
| 16 | (0.3, 0.4] | 11.11 | 0.00 | 10.22   | 0.20  | 11.11    |
|    | (0.4, 0.5] | 11.11 | 0.00 | 9.04    | 1.37  | 5.56     |
|    | (0.5, 0.6] | 11.11 | 0.00 | 8.26    | 0.63  | 11.11    |

Table 7: Per-dataset results on the in-domain mathematical benchmarks. Base, RLVR, and standalone CalibSFT use 6,144 tokens; downstream RL models use 16,384 tokens. Temperature is 0.6, with four rollouts per non-AIME question and 32 per AIME question.

| (a) DeepScaleR-Eval |              |              |              |              | (b) MATH-500 |              |              |              |              | (c) MinervaMath  |              |              |              |              |
|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|--------------|--------------|--------------|--------------|
| Method              | P@1          | AUROC        | Brier        | ECE          | Method       | P@1          | AUROC        | Brier        | ECE          | Method           | P@1          | AUROC        | Brier        | ECE          |
| Base                | 60.84        | 66.31        | 34.09        | 34.06        | Base         | 83.90        | 63.80        | 14.78        | 13.87        | Base             | <b>39.89</b> | <b>63.26</b> | 53.79        | 55.28        |
| RLVR                | <b>70.15</b> | 68.11        | 26.94        | 26.37        | RLVR         | <b>89.15</b> | 67.69        | <b>10.13</b> | <b>8.76</b>  | RLVR             | 37.96        | 60.92        | 57.18        | 58.40        |
| CalibSFT            | 60.88        | <b>73.76</b> | <b>23.60</b> | <b>17.57</b> | CalibSFT     | 83.65        | <b>82.11</b> | 14.70        | 11.53        | CalibSFT         | <b>39.89</b> | 57.50        | <b>30.33</b> | <b>24.20</b> |
| RLCR                | 71.24        | 75.33        | 16.65        | 5.14         | RLCR         | <b>90.50</b> | 76.39        | 7.54         | 7.43         | RLCR             | 36.67        | 66.67        | 35.91        | 38.12        |
| + CalibSFT          | <b>72.47</b> | <b>79.34</b> | <b>15.33</b> | <b>4.58</b>  | + CalibSFT   | 90.15        | <b>89.79</b> | <b>6.56</b>  | <b>3.69</b>  | + CalibSFT       | <b>37.04</b> | <b>76.29</b> | <b>26.81</b> | <b>26.33</b> |
| ReDoubt             | –            | –            | –            | –            | ReDoubt      | –            | –            | –            | –            | ReDoubt          | –            | –            | –            | –            |
| + CalibSFT          | –            | –            | –            | –            | + CalibSFT   | –            | –            | –            | –            | + CalibSFT       | –            | –            | –            | –            |
| CoCA                | 70.16        | 73.17        | 17.73        | 8.51         | CoCA         | <b>89.85</b> | 72.69        | 10.64        | 18.18        | CoCA             | 35.20        | 65.28        | 32.43        | 33.26        |
| + CalibSFT          | <b>71.45</b> | <b>78.71</b> | <b>16.41</b> | <b>7.92</b>  | + CalibSFT   | <b>89.85</b> | <b>87.29</b> | <b>9.09</b>  | <b>15.49</b> | + CalibSFT       | <b>36.95</b> | <b>78.62</b> | <b>23.47</b> | <b>24.42</b> |
| DCPO                | <b>71.45</b> | 71.74        | 17.30        | 6.50         | DCPO         | 89.30        | 70.77        | 7.90         | <b>6.27</b>  | DCPO             | 34.93        | 64.19        | 41.78        | 44.59        |
| + CalibSFT          | 70.46        | <b>77.71</b> | <b>16.06</b> | <b>5.95</b>  | + CalibSFT   | <b>90.20</b> | <b>84.89</b> | <b>7.68</b>  | 7.70         | + CalibSFT       | <b>36.86</b> | <b>74.73</b> | <b>23.33</b> | <b>20.25</b> |
| C3RL                | 70.65        | 62.87        | 24.64        | 25.05        | C3RL         | <b>89.40</b> | 60.22        | <b>9.42</b>  | 9.75         | C3RL             | <b>37.22</b> | 56.00        | 58.42        | 59.49        |
| + CalibSFT          | <b>71.05</b> | <b>71.91</b> | <b>19.93</b> | <b>14.72</b> | + CalibSFT   | 87.70        | <b>84.76</b> | 10.37        | <b>9.36</b>  | + CalibSFT       | 35.85        | <b>77.64</b> | <b>20.14</b> | <b>16.95</b> |
| (d) OlympiadBench   |              |              |              |              | (e) GSM8K    |              |              |              |              | (f) AIME2024     |              |              |              |              |
| Method              | P@1          | AUROC        | Brier        | ECE          | Method       | P@1          | AUROC        | Brier        | ECE          | Method           | P@1          | AUROC        | Brier        | ECE          |
| Base                | 54.41        | 69.82        | 38.93        | 40.13        | Base         | 94.35        | 67.82        | 5.20         | 3.39         | Base             | 48.23        | 81.93        | 39.61        | 42.50        |
| RLVR                | <b>56.31</b> | 72.84        | 39.35        | 39.98        | RLVR         | <b>94.64</b> | 67.97        | <b>5.06</b>  | <b>3.05</b>  | RLVR             | <b>52.92</b> | 82.05        | 39.49        | 41.15        |
| CalibSFT            | 51.67        | <b>79.33</b> | <b>22.48</b> | <b>20.62</b> | CalibSFT     | 94.31        | <b>70.24</b> | 17.90        | 20.78        | CalibSFT         | 45.42        | <b>89.96</b> | <b>11.93</b> | <b>6.03</b>  |
| RLCR                | 57.83        | 81.78        | 19.04        | <b>12.50</b> | RLCR         | <b>94.81</b> | 78.88        | 4.77         | 5.58         | RLCR             | 59.48        | <b>93.59</b> | 13.17        | 19.70        |
| + CalibSFT          | <b>58.46</b> | <b>89.71</b> | <b>16.09</b> | 13.25        | + CalibSFT   | 94.67        | <b>82.60</b> | <b>4.64</b>  | <b>2.72</b>  | + CalibSFT       | <b>64.17</b> | 92.42        | <b>10.97</b> | <b>9.79</b>  |
| ReDoubt             | –            | –            | –            | –            | ReDoubt      | –            | –            | –            | –            | ReDoubt          | –            | –            | –            | –            |
| + CalibSFT          | –            | –            | –            | –            | + CalibSFT   | –            | –            | –            | –            | + CalibSFT       | –            | –            | –            | –            |
| CoCA                | <b>56.38</b> | 80.19        | 20.14        | <b>14.63</b> | CoCA         | <b>94.58</b> | 69.00        | 8.94         | 21.02        | CoCA             | 53.85        | 92.22        | <b>16.27</b> | <b>21.37</b> |
| + CalibSFT          | 56.19        | <b>90.47</b> | <b>18.09</b> | 20.16        | + CalibSFT   | 94.56        | <b>71.85</b> | <b>8.26</b>  | <b>18.10</b> | + CalibSFT       | <b>58.54</b> | <b>95.41</b> | 16.34        | 23.56        |
| DCPO                | 57.05        | 78.43        | 21.85        | 17.75        | DCPO         | <b>94.48</b> | 68.34        | <b>5.50</b>  | 9.13         | DCPO             | 57.92        | 88.84        | 12.32        | <b>8.47</b>  |
| + CalibSFT          | <b>57.46</b> | <b>88.53</b> | <b>15.72</b> | <b>12.14</b> | + CalibSFT   | <b>94.45</b> | <b>73.40</b> | 5.80         | <b>7.48</b>  | + CalibSFT       | <b>58.33</b> | <b>93.89</b> | <b>11.02</b> | 11.85        |
| C3RL                | <b>57.68</b> | 62.66        | 35.85        | 37.52        | C3RL         | <b>94.37</b> | 57.25        | 5.33         | 5.24         | C3RL             | 59.69        | 80.93        | 25.25        | 30.19        |
| + CalibSFT          | 56.64        | <b>84.53</b> | <b>17.91</b> | <b>18.60</b> | + CalibSFT   | 94.29        | <b>64.91</b> | <b>5.26</b>  | <b>5.08</b>  | + CalibSFT       | <b>64.27</b> | <b>85.24</b> | <b>14.04</b> | <b>12.99</b> |
| (g) AIME2025        |              |              |              |              | (h) AIME2026 |              |              |              |              | (i) Overall (ID) |              |              |              |              |
| Method              | P@1          | AUROC        | Brier        | ECE          | Method       | P@1          | AUROC        | Brier        | ECE          | Method           | P@1          | AUROC        | Brier        | ECE          |
| Base                | 35.00        | 82.75        | 48.25        | 52.81        | Base         | 42.50        | 76.08        | 45.13        | 48.07        | Base             | 57.39        | 71.47        | 34.97        | 36.26        |
| RLVR                | <b>43.44</b> | 84.95        | 45.87        | 49.00        | RLVR         | <b>48.23</b> | 80.70        | 43.82        | 45.92        | RLVR             | <b>61.60</b> | 73.16        | 33.48        | 34.08        |
| CalibSFT            | 34.27        | <b>92.48</b> | <b>9.95</b>  | <b>8.02</b>  | CalibSFT     | 39.48        | <b>88.65</b> | <b>13.22</b> | <b>9.79</b>  | CalibSFT         | 56.20        | <b>79.25</b> | <b>18.01</b> | <b>14.82</b> |
| RLCR                | 43.33        | <b>95.53</b> | <b>13.60</b> | 23.06        | RLCR         | 47.81        | <b>94.10</b> | 14.08        | 23.19        | RLCR             | 62.71        | 82.78        | 15.60        | 16.84        |
| + CalibSFT          | <b>48.02</b> | 92.40        | 13.87        | <b>14.28</b> | + CalibSFT   | <b>52.60</b> | 92.48        | <b>13.57</b> | <b>12.04</b> | + CalibSFT       | <b>64.70</b> | <b>86.88</b> | <b>13.48</b> | <b>10.84</b> |
| ReDoubt             | –            | –            | –            | –            | ReDoubt      | –            | –            | –            | –            | ReDoubt          | –            | –            | –            | –            |
| + CalibSFT          | –            | –            | –            | –            | + CalibSFT   | –            | –            | –            | –            | + CalibSFT       | –            | –            | –            | –            |
| CoCA                | 40.31        | <b>95.39</b> | <b>17.04</b> | 29.31        | CoCA         | 48.54        | <b>96.74</b> | <b>14.84</b> | 27.72        | CoCA             | 61.11        | 80.59        | 17.25        | 21.75        |
| + CalibSFT          | <b>45.00</b> | 94.21        | 19.99        | <b>26.43</b> | + CalibSFT   | <b>48.65</b> | 93.96        | 19.52        | <b>25.63</b> | + CalibSFT       | <b>62.65</b> | <b>86.31</b> | <b>16.40</b> | <b>20.21</b> |
| DCPO                | 41.15        | 91.19        | <b>13.33</b> | <b>14.30</b> | DCPO         | 49.58        | 91.78        | <b>11.04</b> | <b>10.41</b> | DCPO             | 61.98        | 78.16        | 16.38        | 14.68        |
| + CalibSFT          | <b>41.25</b> | <b>95.07</b> | 13.81        | 17.73        | + CalibSFT   | <b>50.21</b> | <b>93.07</b> | 12.87        | 14.52        | + CalibSFT       | <b>62.40</b> | <b>85.16</b> | <b>13.29</b> | <b>12.20</b> |
| C3RL                | <b>43.96</b> | 79.01        | 36.25        | 42.69        | C3RL         | 52.81        | 84.29        | 27.21        | 33.72        | C3RL             | 63.22        | 67.90        | 27.80        | 30.45        |
| + CalibSFT          | 43.12        | <b>86.74</b> | <b>21.86</b> | <b>25.49</b> | + CalibSFT   | <b>53.65</b> | <b>91.31</b> | <b>15.69</b> | <b>18.30</b> | + CalibSFT       | <b>63.32</b> | <b>80.88</b> | <b>15.65</b> | <b>15.18</b> |

Table 8: Per-dataset results on the revised eight-dataset out-of-domain suite. Each question has four sampled responses at temperature 0.6. WebQuestions contains only examples with a single gold answer.

| (a) HotpotQA |              |              |              |              | (b) TriviaQA           |              |              |              |              | (c) DROP          |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|------------------------|--------------|--------------|--------------|--------------|-------------------|--------------|--------------|--------------|--------------|
| Method       | P@1          | AUROC        | Brier        | ECE          | Method                 | P@1          | AUROC        | Brier        | ECE          | Method            | P@1          | AUROC        | Brier        | ECE          |
| Base         | 64.68        | <b>57.68</b> | 32.82        | 32.16        | Base                   | <b>55.24</b> | 75.80        | 35.68        | 36.93        | Base              | 80.36        | 59.13        | 18.22        | 17.03        |
| RLVR         | 64.58        | 54.16        | 33.10        | 32.17        | RLVR                   | 54.97        | <b>76.01</b> | 40.59        | 41.40        | RLVR              | <b>83.53</b> | 59.32        | <b>15.34</b> | <b>13.31</b> |
| CalibSFT     | <b>65.70</b> | 54.18        | <b>29.38</b> | <b>20.57</b> | CalibSFT               | 55.21        | 65.05        | <b>28.44</b> | <b>20.98</b> | CalibSFT          | 79.92        | <b>59.38</b> | 31.60        | 28.93        |
| RLCR         | 65.80        | 59.10        | 25.97        | 19.77        | RLCR                   | 55.24        | <b>81.66</b> | 23.84        | 23.30        | RLCR              | 81.44        | 63.34        | <b>14.72</b> | <b>5.43</b>  |
| + CalibSFT   | <b>66.27</b> | <b>59.23</b> | <b>24.25</b> | <b>10.85</b> | + CalibSFT             | <b>55.25</b> | 81.53        | <b>18.89</b> | <b>10.71</b> | + CalibSFT        | <b>82.32</b> | <b>74.44</b> | 14.99        | 10.75        |
| ReDoubt      | –            | –            | –            | –            | ReDoubt                | –            | –            | –            | –            | ReDoubt           | –            | –            | –            | –            |
| + CalibSFT   | –            | –            | –            | –            | + CalibSFT             | –            | –            | –            | –            | + CalibSFT        | –            | –            | –            | –            |
| CoCA         | 64.90        | 56.43        | <b>22.77</b> | <b>6.02</b>  | CoCA                   | 54.77        | 77.62        | 21.63        | <b>12.97</b> | CoCA              | 82.47        | 61.58        | <b>15.03</b> | <b>11.56</b> |
| + CalibSFT   | <b>66.10</b> | <b>61.57</b> | 22.92        | 11.39        | + CalibSFT             | <b>54.89</b> | <b>81.85</b> | <b>20.98</b> | 13.16        | + CalibSFT        | <b>82.50</b> | <b>73.91</b> | 17.95        | 21.99        |
| DCPO         | <b>66.15</b> | 57.88        | 25.33        | 18.77        | DCPO                   | 54.55        | 79.86        | 25.93        | 23.91        | DCPO              | <b>83.45</b> | 59.34        | <b>13.25</b> | <b>2.30</b>  |
| + CalibSFT   | 65.97        | <b>60.70</b> | <b>24.43</b> | <b>14.92</b> | + CalibSFT             | <b>55.12</b> | <b>82.56</b> | <b>17.59</b> | <b>8.25</b>  | + CalibSFT        | 80.42        | <b>75.92</b> | 16.54        | 13.64        |
| C3RL         | 65.38        | 52.47        | 33.39        | 33.12        | C3RL                   | 53.76        | <b>73.01</b> | 35.61        | 38.53        | C3RL              | <b>82.65</b> | 54.56        | <b>16.37</b> | <b>16.07</b> |
| + CalibSFT   | <b>66.50</b> | <b>58.94</b> | <b>28.45</b> | <b>19.78</b> | + CalibSFT             | <b>54.11</b> | 71.36        | <b>22.75</b> | <b>12.50</b> | + CalibSFT        | 77.44        | <b>64.78</b> | 23.65        | 22.03        |
| (d) MuSiQue  |              |              |              |              | (e) LiveBenchReasoning |              |              |              |              | (f) NQOpen        |              |              |              |              |
| Method       | P@1          | AUROC        | Brier        | ECE          | Method                 | P@1          | AUROC        | Brier        | ECE          | Method            | P@1          | AUROC        | Brier        | ECE          |
| Base         | 45.84        | <b>62.90</b> | 41.91        | 42.33        | Base                   | 52.88        | 61.81        | 40.53        | 40.59        | Base              | 21.12        | 66.31        | 66.04        | 70.59        |
| RLVR         | 47.18        | 58.12        | 46.19        | 46.75        | RLVR                   | 29.25        | 60.32        | 61.75        | 64.59        | RLVR              | <b>24.13</b> | <b>67.07</b> | 68.92        | 71.55        |
| CalibSFT     | <b>47.19</b> | 62.25        | <b>26.51</b> | <b>13.95</b> | CalibSFT               | <b>53.00</b> | <b>67.67</b> | <b>25.40</b> | <b>17.74</b> | CalibSFT          | 21.64        | 59.59        | <b>27.33</b> | <b>29.16</b> |
| RLCR         | 47.77        | 66.21        | 25.68        | 15.35        | RLCR                   | 31.75        | 68.00        | 28.49        | 29.09        | RLCR              | 23.32        | 70.48        | 43.84        | 52.43        |
| + CalibSFT   | <b>49.97</b> | <b>67.87</b> | <b>23.22</b> | <b>6.13</b>  | + CalibSFT             | <b>42.25</b> | <b>68.41</b> | <b>21.94</b> | <b>7.75</b>  | + CalibSFT        | <b>23.80</b> | <b>72.24</b> | <b>26.44</b> | <b>30.89</b> |
| ReDoubt      | –            | –            | –            | –            | ReDoubt                | –            | –            | –            | –            | ReDoubt           | –            | –            | –            | –            |
| + CalibSFT   | –            | –            | –            | –            | + CalibSFT             | –            | –            | –            | –            | + CalibSFT        | –            | –            | –            | –            |
| CoCA         | 46.69        | <b>67.03</b> | 23.69        | 10.18        | CoCA                   | 32.12        | 66.75        | 25.79        | 23.64        | CoCA              | 22.76        | 66.59        | 35.28        | 43.26        |
| + CalibSFT   | <b>48.76</b> | 66.88        | <b>23.56</b> | <b>7.72</b>  | + CalibSFT             | <b>41.50</b> | <b>68.03</b> | <b>23.24</b> | <b>7.96</b>  | + CalibSFT        | <b>23.72</b> | <b>70.77</b> | <b>24.86</b> | <b>28.69</b> |
| DCPO         | 47.34        | 65.48        | 28.28        | 20.53        | DCPO                   | 31.13        | 67.14        | 32.10        | 33.12        | DCPO              | <b>23.69</b> | 69.80        | 46.27        | 53.69        |
| + CalibSFT   | <b>48.96</b> | <b>70.49</b> | <b>22.75</b> | <b>8.55</b>  | + CalibSFT             | <b>37.75</b> | <b>71.77</b> | <b>21.12</b> | <b>8.83</b>  | + CalibSFT        | 23.58        | <b>73.13</b> | <b>22.65</b> | <b>25.97</b> |
| C3RL         | 45.55        | 63.39        | 39.88        | 40.04        | C3RL                   | 33.00        | <b>57.18</b> | 53.55        | 55.46        | C3RL              | <b>22.43</b> | <b>64.77</b> | 63.63        | 68.49        |
| + CalibSFT   | <b>47.47</b> | <b>66.39</b> | <b>24.46</b> | <b>11.87</b> | + CalibSFT             | <b>44.62</b> | 53.33        | <b>28.11</b> | <b>22.78</b> | + CalibSFT        | 22.08        | 61.18        | <b>22.25</b> | <b>17.54</b> |
| (g) PopQA    |              |              |              |              | (h) WebQuestions       |              |              |              |              | (i) Overall (OOD) |              |              |              |              |
| Method       | P@1          | AUROC        | Brier        | ECE          | Method                 | P@1          | AUROC        | Brier        | ECE          | Method            | P@1          | AUROC        | Brier        | ECE          |
| Base         | 20.85        | <b>74.94</b> | 57.89        | 64.10        | Base                   | 21.88        | <b>67.10</b> | 68.16        | 71.80        | Base              | 45.36        | <b>65.71</b> | 45.15        | 46.94        |
| RLVR         | <b>21.37</b> | 72.78        | 68.14        | 71.96        | RLVR                   | <b>26.02</b> | 65.76        | 68.29        | 70.41        | RLVR              | 43.88        | 64.19        | 50.29        | 51.52        |
| CalibSFT     | 21.00        | 63.88        | <b>24.80</b> | <b>26.68</b> | CalibSFT               | 23.59        | 59.02        | <b>32.52</b> | <b>35.25</b> | CalibSFT          | <b>45.91</b> | 61.38        | <b>28.25</b> | <b>24.16</b> |
| RLCR         | <b>21.29</b> | <b>76.88</b> | 37.71        | 47.90        | RLCR                   | <b>25.07</b> | 69.40        | 48.86        | 56.11        | RLCR              | 43.96        | 69.38        | 31.14        | 31.17        |
| + CalibSFT   | 21.16        | 76.55        | <b>24.76</b> | <b>32.22</b> | + CalibSFT             | 24.63        | <b>70.20</b> | <b>31.41</b> | <b>36.33</b> | + CalibSFT        | <b>45.71</b> | <b>71.31</b> | <b>23.24</b> | <b>18.20</b> |
| ReDoubt      | –            | –            | –            | –            | ReDoubt                | –            | –            | –            | –            | ReDoubt           | –            | –            | –            | –            |
| + CalibSFT   | –            | –            | –            | –            | + CalibSFT             | –            | –            | –            | –            | + CalibSFT        | –            | –            | –            | –            |
| CoCA         | <b>21.06</b> | 73.92        | 30.84        | 39.88        | CoCA                   | 24.65        | 64.40        | 38.35        | 45.46        | CoCA              | 43.68        | 66.79        | 26.67        | 24.12        |
| + CalibSFT   | 20.88        | <b>76.99</b> | <b>24.12</b> | <b>31.76</b> | + CalibSFT             | <b>26.06</b> | <b>69.71</b> | <b>25.93</b> | <b>28.47</b> | + CalibSFT        | <b>45.55</b> | <b>71.21</b> | <b>22.95</b> | <b>18.89</b> |
| DCPO         | 20.99        | <b>77.02</b> | 41.10        | 49.96        | DCPO                   | 24.59        | 68.48        | 50.52        | 57.04        | DCPO              | 43.99        | 68.13        | 32.85        | 32.41        |
| + CalibSFT   | <b>21.18</b> | 76.52        | <b>21.29</b> | <b>28.02</b> | + CalibSFT             | <b>25.61</b> | <b>70.87</b> | <b>27.45</b> | <b>30.32</b> | + CalibSFT        | <b>44.82</b> | <b>72.75</b> | <b>21.73</b> | <b>17.31</b> |
| C3RL         | <b>20.81</b> | <b>71.98</b> | 54.60        | 61.92        | C3RL                   | <b>23.94</b> | 60.19        | 67.27        | 70.35        | C3RL              | 43.44        | 62.19        | 45.54        | 48.00        |
| + CalibSFT   | 20.66        | 66.98        | <b>16.56</b> | <b>12.95</b> | + CalibSFT             | <b>23.94</b> | <b>61.29</b> | <b>24.87</b> | <b>19.55</b> | + CalibSFT        | <b>44.60</b> | <b>63.03</b> | <b>23.89</b> | <b>17.38</b> |

rubric-based LLM judgments (LoVeC) (Zhang et al., 2026). For a fair comparison, we construct the training datasets from a shared response pool. SaySelf uses its correct responses, following its SFT setup. We sample questions uniformly across empirical success rates so that the comparison covers both easier and harder questions.

For each question in DeepScaleR-Train, we use 10 responses generated by Qwen3-8B. With  $z_{g,i} \in \{0, 1\}$  denoting the response correctness, the empirical success rate is  $q_g = \frac{1}{10} \sum_{i=1}^{10} z_{g,i}$ . We select 300 questions at each  $q_g \in \{0.1, 0.2, \dots, 0.9\}$  and retain one correct and one incorrect response per question. Finally, We have 5,400 responses from 2,700 questions, with equal numbers of correct and incorrect responses at each success rate.

On this shared training subset, we construct the five confidence targets as follows:

- **CSFT.** Both responses receive the group success rate,  $c_{g,i}^* = q_g$ .
- **UaIT.** The target is  $c_{g,i}^* = \exp(-\ell_{g,i})$ , where  $\ell_{g,i}$  is the mean negative log-likelihood of the sampled completion tokens under the generating model.
- **SaySelf.** SaySelf uses an embedding model to group semantically similar responses and derives confidence from cluster frequency. We embed the ten responses to each question using Qwen3-Embedding-8B and cluster them with a cosine similarity threshold of 0.98. Since SaySelf uses only correct responses for SFT, we retain the 2,700 selected correct responses and assign each the frequency of its cluster,  $c_{g,i}^* = |C_{g,i}|/10$ .
- **LoVeC.** LoVeC uses an LLM judge to assess the factuality of generated statements and supervise their confidence estimates. Our rubric assigns 10 to correct answers with sound reasoning, 8–9 to correct answers with minor reasoning gaps, 5–7 to partially correct or ambiguous responses, 2–4 to meaningful but incorrect attempts, and 0–1 to invalid or clearly wrong responses. We normalize its score  $s_{g,i} \in \{0, \dots, 10\}$  to obtain  $c_{g,i}^* = s_{g,i}/10$ .
- **CalibSFT.** Follow our official implementation, the target combines group success rate with response correctness,  $c_{g,i}^* = 0.5q_g + 0.5z_{g,i}$ .

All variants initialize from Qwen3-8B and use the same supervision objective. Reasoning and answer tokens receive supervision on correct responses, while confidence tokens receive supervision on all responses. We train for 80 steps (40 for SaySelf) with a batch size of 256 and a learning rate of  $2 \times 10^{-6}$ . We report the proportion of training targets in each of 10 equal-width confidence intervals for all five methods in Table 9.

### D.3 TRAINING STABILITY

**[TODO:]** Show training curves and report the mean and standard deviation of evaluation metrics across training seeds.

### D.4 ADDITIONAL METRICS

**[TODO:]** Report the additional calibration and selective-prediction metrics, including MacroCE, adaptive ECE, PCE, and AURC/EAURC, together with their definitions and aggregation protocols. Include ECE sensitivity to 10, 15, and 20 bins, and plots comparing calibration metrics with AUROC and Pass@1.

### D.5 GENERALIZATION TO ADDITIONAL MODELS

**[TODO:]** Evaluate CalibSFT and representative downstream RL objectives with OLMo and Gemma, using the same ID/OOD protocol as the main experiment.

D.6 CONFIDENCE VISUALIZATION

**[TODO:]** Show reliability diagrams and confidence distributions for models with and without CalibSFT initialization.

D.7 CONFIDENCE DISTRIBUTION ANALYSIS

**[TODO:]** Report Top-1 and Top-3 confidence mass and confidence entropy, with per-dataset results and ID/OOD averages.

D.8 TRAINING EFFICIENCY

**[TODO:]** Compare CalibSFT and confidence-aware RL using wall-clock time, GPU-hours, optimization steps, and generated rollout/token counts.

D.9 QUALITATIVE CASE STUDIES

**[TODO:]** Include representative high-confidence errors, low-confidence correct answers, confidence corrections preserved after RL, and failure cases.